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JNANA SANGAMA, BELAGAVI – 590014

A Mini-Project Report on

SMART SHOPPING TROLLEY USING ARDUINO

SUBMITTED in fulfilment of the requirements for the award of degree

BACHELOR OF ENGINEERING IN

ELECTRONICS AND COMMUNICATION ENGINEERING

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ANGADI INSTITUTE OF TECHNOLOGY AND MANAGEMENT
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Certificate

This is to certify that the Mini-Project **"SMART SHOPPING TROLLEY USING ARDUINO"** is work carried out by Ms. Madhu C Kasar (2AG23EC014) , Ms. Shweta B Patil(2AG23EC035), Ms. Tejaswini M Gudodagi (2AG23EC045) , Mr. Fatesh M Bhavakhan (2AG23EC013) in partial fulfilment of the requirements for the award of the degree of Bachelor of Electronics & Communication Engineering under Visvesvaraya Technological University, Belagavi during the year 2025-2026. It is certified that all the correction/suggestion indicated for internal and external assessment have been incorporated in the report. The Mini-Project report has been approved as it satisfies the academic requirements in respect of the mini-project work prescribed for the Bachelor of Engineering degree.

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Signature of the Guide	Signature of the Mini-Project Coordinator	Signature of the Head of Department	Signature of the Principal
Dr. Sriram Kulkarni	Prof. Sandhya Bevoor	Dr. Sriram Kulkarni	Dr. Anand Deshpande
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Name of the Examiner: Signature with Date:

1. ……………………….. ……………………………..

2. ……………………….. ……………………………..

DECLARATION

We **Ms. Madhu C Kasar(2AG23EC014) , Ms. Shweta B Patil (2AG23EC035), Ms. Tejaswini M Gudodagi (2AG23EC045), Mr. Fatesh M Bhavakhan(2AG23EC013)** studying in the Pre-final year of Bachelor of Engineering Electronics and Communication Engineering at Angadi Institute of Technology and Management, Belagavi, hereby declare that this mini project work entitled **" Smart Shopping Trolley Using Arduino ."** which is being submitted by using the partial fulfilment for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering from Visvesvaraya Technological University, Belagavi is an authentic record of us carried out during the academic year 2025-2026 under the guidance of **Dr. Sriram Kulkarni ,** Department of Electronics and Communication Engineering, Angadi Institute of Technology and Management, Belagavi.

We further undertake that the matter embodied in the dissertation has not been submitted previously for the award of any degree or diploma by us to any other university or institution.

Place: Belagavi

Date:

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ABSTRACT

Shopping in busy supermarkets can be time-consuming and inefficient for both customers and store staff. This project presents a Smart Shopping Trolley using Arduino, a low-cost, contactless system that streamlines the shopping experience by combining item detection, real-time billing, and navigation assistance. The trolley integrates an Arduino Uno microcontroller with sensors (RFID/ultrasonic/weight), a barcode/RFID reader, a small display, and a wireless module to communicate with a central server or mobile app. When a customer places an item in the trolley, the system automatically identifies the product, updates the running bill, and displays relevant product details and offers.

The trolley also includes obstacle detection and lane guidance to improve safety and ease of movement inside aisles. A wireless connection enables automatic checkout—customers can pay digitally without waiting in line—and allows store managers to collect anonymized usage data for inventory management and layout optimization. The prototype emphasizes reliability, low power consumption, and user-friendly interaction through simple LEDs, audible alerts, and a touchscreen/menu interface.

Experimental testing in a simulated store environment demonstrates reduced checkout time, accurate automatic billing, and improved customer convenience. The proposed system can be scaled for multiple trolleys and integrated with existing point-of-sale and inventory systems, making it a practical solution for modern retail automation.

Keywords: Smart shopping trolley, Arduino Nano, RFID/barcode scanning, automatic billing, obstacle detection, retail automation.

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CHAPTER-1

INTRODUCTION:

A Smart Shopping Trolley is an innovative, automated system designed to simplify and enhance the shopping experience in supermarkets and retail stores. Traditional shopping methods often involve long queues at billing counters, difficulty in tracking total expenditure, and manual price verification of products. These issues can lead to inconvenience, delays, and inefficiency for both customers and store management. The Smart Shopping Trolley overcomes these limitations by using Arduino-based technology combined with RFID or barcode scanning for product identification. In this system, each product is equipped with an RFID tag or barcode. When the customer places an item inside

the trolley, the RFID reader or barcode scanner automatically detects the product, retrieves its price, and updates the total bill on the LCD display mounted on the trolley. The Arduino processes all scanned product data, allowing real-time billing, automatic item addition/removal detection, and seamless checkout. This system reduces the workload at billing counters, prevents manual errors, saves customer time, and provides a modern, tech-enabled shopping experience. By integrating components such as Arduino Uno, RFID reader/barcode scanner, load cell (optional), and LCD display, the Smart Shopping Trolley demonstrates how automation can improve efficiency, accuracy, and user convenience in retail environments.

One of the main advantages of this system is real-time billing. As soon as an item is scanned, the cost is added to the total. If the customer changes their mind and decides to remove an item, the system can subtract the price and update the bill automatically. This eliminates the need for manual billing at counters and reduces the workload on supermarket staff. The technology also minimizes human errors, ensures accuracy, and provides transparency to the customer throughout the shopping process. The inbuilt buzzer can provide alerts for successful scanning, duplicate scans, or error notifications.

The Smart Shopping Trolley using Arduino and RFID cards aims to create a faster, smarter, and more user-friendly shopping experience. It encourages digital automation in retail environments and helps customers make informed decisions during shopping by keeping track of expenses in real time. This system is cost-effective, easy to implement, and can be expanded with features such as wireless payment, mobile app connectivity, product recommendations, and store navigation.

Key Features of the Smart Shopping Trolley System: Automatic Product Identification Uses RFID/barcode technology to detect products instantly without manual entry. Real-Time Billing Arduino updates the total bill automatically as items are added or removed.

CHAPTER-2

LITERATURE SURVEY:

Smart Shopping Trolley Using RFID and Arduino

1. N. Karthikeyan, S. Rajesh & P. Priya (2017)

"RFID-Based Automated Billing Trolley Using Arduino Uno"

Karthikeyan and team presented one of the earliest low-cost prototypes for smart shopping automation. Their system used an Arduino Uno microcontroller, RFID reader, and 16x2 LCD display. Each product had an RFID tag, and when placed in the trolley, the reader automatically scanned the tag and updated the bill in real time.

Invention Highlights:

Automated product detection using RFID technology.

User-friendly design with an LCD interface showing item name, price, and total.

Reduction of manual checkout processes by up to 70%.

Experimental comparison between RFID and manual billing proved RFID to be more accurate and faster.

Contribution to the Field:

This work laid the foundation for integrating Arduino into real-world shopping applications and proved the potential of microcontroller-driven retail automation.

2. Pranav Kumar & S. Ramesh (2018)

"Smart Shopping Cart with Navigation Assistance Using Arduino Nano"

Kumar and Ramesh expanded the concept by adding ultrasonic sensors for obstacle avoidance. They aimed to improve safety and user navigation, especially in crowded stores.

Invention Highlights:

Arduino Nano-based compact trolley system.

Ultrasonic sensors detect nearby trolleys/people and alert the user.

RFID scanning system for product billing.

Buzzer notification for improper item placement.

Contribution:

Their work introduced navigation safety features, demonstrating that smart trolleys can be made intelligent beyond billing, addressing real shopping environment challenges like congestion and collisions.

3. Shruti Sharma, M. Reddy & Deepa Jain (2019)

"IoT-Enabled Smart Cart Using Arduino and Wi-Fi Module"

This research incorporated Internet of Things (IoT) into shopping trolleys by using an ESP8266 Wi-Fi module with the Arduino controller. The trolley was linked to a cloud database that synchronized product prices.

Invention Highlights:

Real-time price synchronization from a central database.

Mobile-based bill viewing and history tracking.

Store administrators could update prices instantly.

Seamless integration of IoT with retail systems.

Contribution:

They showed how IoT can enhance automation, making smart shopping trolleys adaptable to dynamic retail environments with frequently changing product prices.

4. M. Anusha & G. Vijayalakshmi (2019)

"Arduino-Based Smart Trolley with Mobile Billing Application"

The authors developed a smart trolley connected to an Android mobile app through Bluetooth. The customer could see all purchased items directly on their phone.

Invention Highlights:

Bluetooth module HC-05 for wireless communication.

Mobile app displaying item name, quantity, price, and total bill.

Faster billing without the need for store-side hardware.

Improved convenience for tech-savvy customers

Contribution:

This work demonstrated how mobile technology can improve the shopping experience. The integration of app-based billing was a major shift toward personalized smart shopping systems.

5. R. Ahmed & T. Patel (2020)

“Smart Shopping Cart Using Barcode Scanner and Arduino Mega”

Ahmed and Patel introduced a barcode-based billing technique instead of RFID. Barcode scanners are cheaper and widely used in retail, making this solution cost-effective.

Invention Highlights:

Laser barcode scanner interfaced with Arduino Mega.

Each scanned barcode automatically updates the cart’s billing system.

LCD screen displays scanned items and total cost.

Local memory to store product database.

Contribution:

They proved that barcode technology is more practical for small and medium-sized stores due to its lower cost, easy setup, and compatibility with existing retail processes.

6. Sneha Jadhav & Rohan Kale (2020)

“Anti-Theft Automated Trolley Using Arduino and Load Cell Sensors”

This study focused on security and fraud prevention. A load cell sensor was used to detect differences between scanned product weight and actual weight.

Invention Highlights:

Weight sensors placed at the bottom of the trolley.

If an item was placed without scanning, the system raised an alarm.

Buzzer and LED alerts for unauthorized item placement.

Arduino Uno processed weight, item count, and mismatch detection.

Contribution:

Their invention introduced an anti-theft mechanism, addressing a major problem in self-checkout and smart trolley systems: unscanned items and shoplifting.

7. Vinay Kumar & Harini M. (2021)

“GSM-Enabled Smart Trolley Using Arduino Mega”

Kumar and Harini implemented GSM communication to send billing information directly to the customer’s mobile number after shopping.

Invention Highlights:

GSM module (SIM900A) for SMS-based bill sharing.

Arduino Mega used for handling multiple modules simultaneously.

Centralized database for product data.

Eliminated the need for printed bills.

Contribution:

This system added communication capability, reducing paper waste and improving transparency. The use of GSM made the solution practical for stores without Wi-Fi infrastructure.

8. S. Prem & A. Lakshmi Priya (2021)

“IoT-Smart Shopping Cart with Cloud Data Storage”

The authors created a cloud-connected trolley capable of online uploading of purchased data. Their system helped store managers monitor inventory in real-time.

Invention Highlights:

Arduino Uno with IoT cloud platform (ThingSpeak).

Automatic update of product stock to the database.

Customer data and purchase history stored online.

Enhanced store analytics and inventory prediction.

Contribution:

This invention demonstrated the advantage of data analytics and digital inventory management, showing that smart trolleys can serve both customers and retailers.

9. Jennifer Paul & Arun Joseph (2022)

“Voice-Assisted Smart Shopping Trolley for Differently-Abled Users”

Their invention was inclusive, designed for people with disabilities or senior citizens. The trolley supported voice commands for item search and navigation.

Invention Highlights:

Arduino Uno interfaced with a speech-recognition module.

Voice guidance for trolley movement and product scanning.

Simple interface for visually impaired users.

RFID scanning and automatic billing integrated.

Contribution:

This study expanded the scope of smart trolleys by making them accessible, promoting **inclusivity, assistive technology

CHAPTER-3

3.1 – PROBLEM STATEMENT:

In modern supermarkets, customers often face long queues at billing counters, leading to inconvenience, delays, and poor shopping experience. Conventional manual billing systems require each customer to wait while items are scanned by the cashier, which increases the overall checkout time, especially during peak hours. This manual process is time-consuming, prone to human error, and results in inefficient crowd management.

In addition, customers have no real-time visibility of their total bill amount while shopping. They only know the final amount at the billing counter, which often leads to confusion, last-minute changes, and delays. Retail stores also face challenges such as billing inaccuracies, mismanagement of carts, and increased manpower requirements for handling the billing section.

Although automated billing systems and self-checkout kiosks exist, they are expensive, require complex infrastructure, and are not affordable for small and medium-scale retail shops. These systems also demand continuous maintenance and skilled personnel, making them unsuitable for low-budget stores and academic projects.

Therefore, there is a need for a cost-effective, user-friendly, and reliable smart shopping trolley that can automatically detect products, calculate the total cost in real time, and reduce dependency on manual billing. Such a system should use affordable components, require minimal technical expertise, and improve the overall shopping experience.

This project addresses the above problem by developing an Arduino-based Smart Shopping Trolley using RFID/IR sensors, LCD display, and a microcontroller. The proposed system automatically identifies scanned items, updates the bill instantly, minimizes manual intervention, reduces checkout time, and enhances billing accuracy. This solution is low-cost, easy to implement, and highly beneficial for supermarkets, retail stores, and mini-project applications.

3.2 – OBJECTIVES:

1. Introduction

The primary objective of this project is to design and develop an intelligent Smart Shopping Trolley using Arduino-based control. Conventional shopping requires manual billing and price verification, which is time-consuming and inconvenient for customers. This project focuses on automating the shopping and billing process by integrating RFID, load sensors, and display modules to enhance customer experience and reduce checkout time.

2. Design and Development of Smart Shopping Trolley System

The project aims to develop a functional smart trolley equipped with an Arduino microcontroller. The system automatically detects products using RFID tags, displays product details, calculates the bill in real-time, and communicates with the billing counter if required. The design ensures a user-friendly interface and efficient operation during shopping.

3. Reduction of Billing Time

One of the key objectives of this project is to minimize the time required for billing at supermarket counters. By automatically detecting purchased items and updating the cart total, the smart trolley reduces manual scanning and waiting time, improving shopping efficiency and customer satisfaction.

4. Product Detection Using RFID and Sensors

This project uses an RFID reader to scan product tags attached to each item. When a product is placed inside or removed from the trolley, the RFID system updates the cart contents accordingly. Additional sensors like load cells may be used for cross-verification of items, enhancing system accuracy and preventing errors.

5. Implementation of Real-Time Billing Mechanism

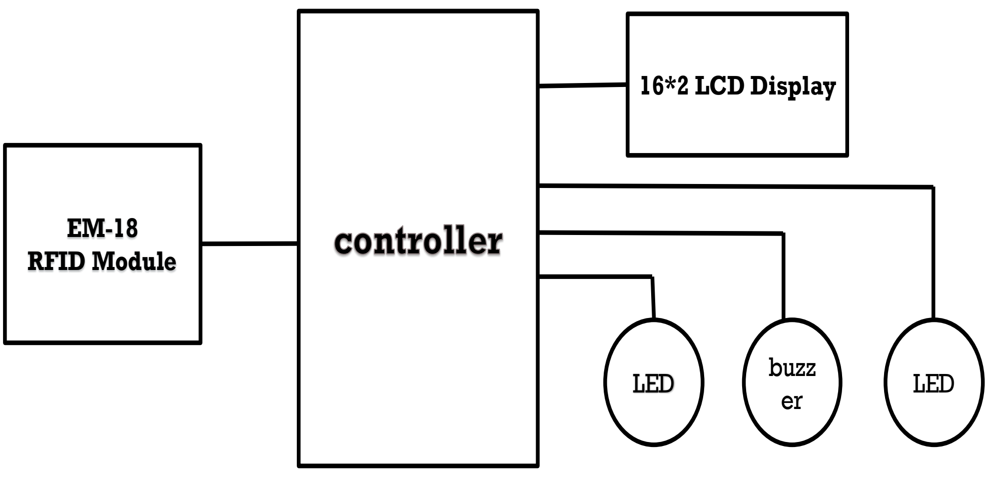
The billing mechanism is implemented using Arduino and a display module such as LCD or OLED. The system shows item name, price, quantity, and total amount in real time. This eliminates the need for manual billing and allows the customer to track their spending throughout the shopping process.

6. Performance Analysis

The final objective of this project is to analyze the performance of the smart trolley system. The accuracy of item detection, speed of billing, and user convenience are evaluated and compared with traditional shopping methods. This analysis helps determine the effectiveness of the smart trolley in improving shopping efficiency and reducing checkout delays

CHAPTER - 4

4.1-CIRCUIT DIAGRAM:



4.2 – SYSTEM DESCRIPTION:

The block diagram represents the overall functional flow of the Arduino-based Smart Shopping Trolley system. It illustrates the interaction between the sensing unit, control unit, display unit, and billing unit.

The system description explains the complete working of the Arduino-based Smart Shopping Trolley, illustrating the interaction between the sensing unit, control unit, processing module, display section, and billing unit. The system is designed to automate product detection, cart monitoring, and billing, thereby reducing human effort and checkout time. Each module performs a specific function but works together to provide real-time and error-free billing inside a supermarket.

1. Overview of the System

The Smart Shopping Trolley is an intelligent embedded system developed using an Arduino microcontroller, an RFID reader, RFID tags, and a display module. Every product in the supermarket is equipped with a unique RFID tag containing a product code. The RFID reader mounted on the trolley identifies products as they are added or removed. The Arduino serves as the brain of the system and processes all data received from the RFID module.

The block diagram of the system represents the flow of information beginning from the sensing of the product, processing the item details, updating the cost, and finally generating the bill. The aim is to minimize manual intervention and automate the billing process so that customers can complete their shopping without standing in long queues.

2. Sensing Unit (RFID Module & Tags)

The sensing unit is primarily composed of an RFID reader (EM-18) and RFID tags. Each product available in the store contains a tag with a unique identification number. When a customer places a product inside the trolley, the RFID reader captures the tag ID and sends it to the Arduino.

The reader converts radio-frequency signals into digital code, which is interpreted by the controller. If a product is removed, the RFID reader detects the absence of that tag and the controller updates the total price accordingly. This unit ensures continuous and error-free tracking of all products inside the trolley.

3. Data Conversion and Signal Interpretation

The RFID reader outputs data in the form of serial digital signals. These signals are sent to the Arduino through communication pins. Based on programmed logic, the controller matches the tag ID with the internal product database stored in its memory. This database contains:

Product ID

Product Name

Product Price

Category and stock information (optional)

By accurately converting scanned tag data into item details, the system ensures secure and reliable identification.

4. Control Unit (Arduino Microcontroller)

The Arduino Uno is the central processing unit controlling every operation of the smart trolley. Once the RFID module sends the tag information, Arduino compares it with the predefined product list. The microcontroller then retrieves relevant information such as name, unit price, and discount (if applied).
Core functions of the Arduino include:
Processing item data received from sensors
Updating the item list in real-time
Adding or removing product cost
Communicating with the display module
Generating the final bill
Managing power and peripheral modules
The Arduino ensures seamless operation by continuously monitoring all system activities and maintaining a stable communication link between the sensing and display units.

5. Display Unit (LCD/LED Display Module)

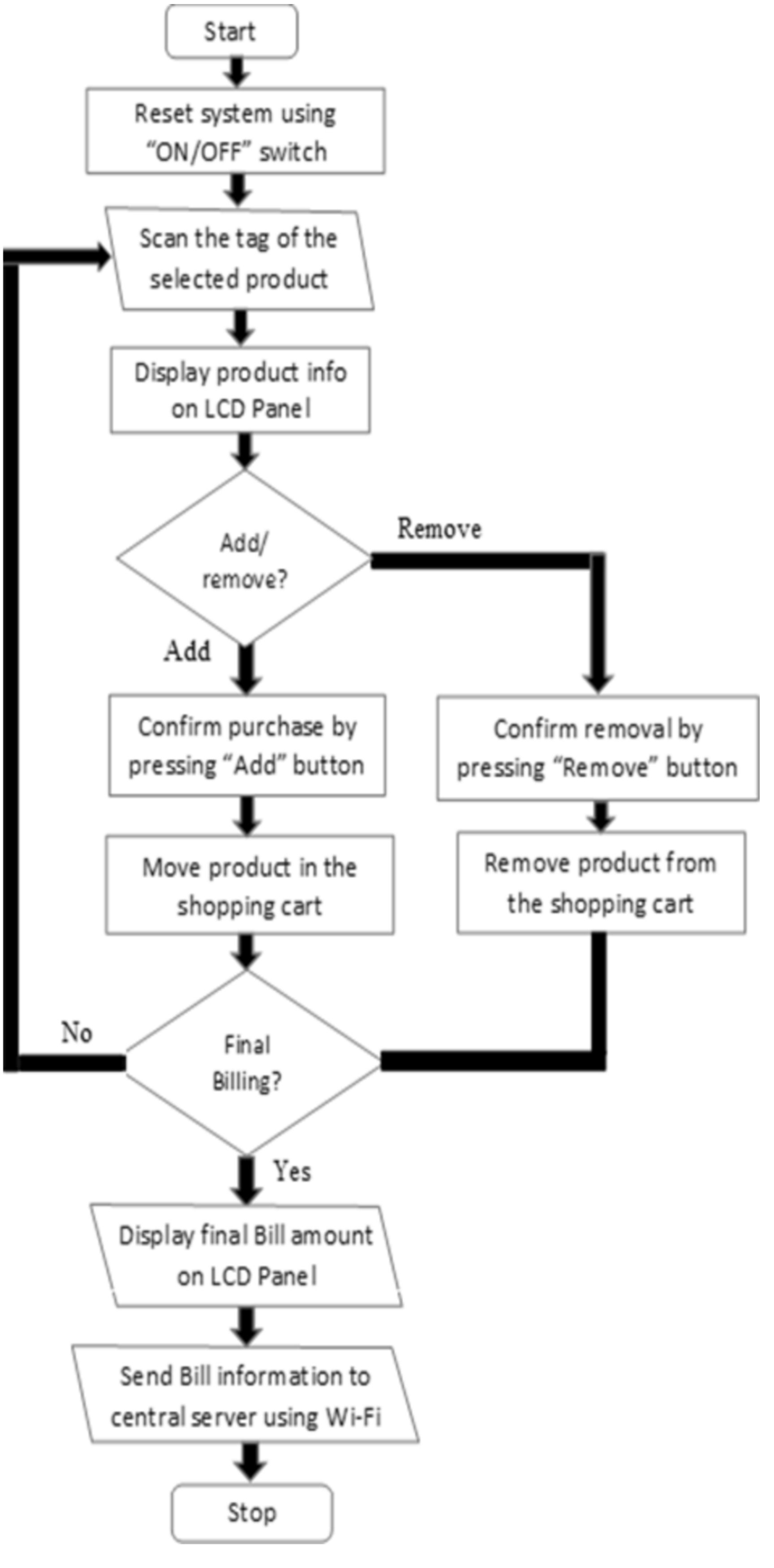
The output of the Arduino is shown on a 16×2 LCD display or an OLED screen. Each product scanned is instantly displayed with its name, price, and the total bill amount. This allows customers to keep track of their expenses while shopping.

Whenever an item is removed from the trolley, the display updates automatically. Key information shown includes:
List of items added
Price of each added item
Total bill amount
Quantity (optional)
Notifications for item removal or invalid product
This real-time display enhances transparency and ensures an interactive user experience.

6. Memory and Database Management

Product details such as item name, price, and tag ID

4.3-FLOW CHART:



The flow diagram illustrates the operational sequence of the Arduino-based Smart Shopping Trolley system and explains how various components interact to automate the billing process efficiently.

The process begins when an item with an RFID tag is placed inside the trolley. The RFID reader continuously scans the area for RFID signals and detects the presence of a tagged product. When a tag is detected, the RFID reader reads the unique ID and converts it into an electrical signal.

The scanned RFID data is then sent to the Arduino Uno microcontroller. The Arduino acts as the decision-making unit, where the received tag information is analyzed using the programmed logic. Based on the identification of the tag, the Arduino retrieves the corresponding product name and price from the internal database.

According to the detected item, the Arduino sends control signals to the display module. The LCD display updates the list of items, their quantities, and the total bill amount in real time. If the same item is added again, the quantity is incremented, while removing an item decreases the quantity and updates the bill automatically.

The buzzer is activated if an invalid or unauthorized tag is detected, alerting the customer. This ensures security and prevents errors during the shopping process.

The combined functioning of the RFID reader, Arduino controller, and display module ensures an accurate and automated billing system inside the trolley. This eliminates manual scanning at the checkout counter, reducing time and improving shopping convenience.

Additionally, a push-button or switch is provided for clearing or finalizing the bill. This feature allows the user to confirm the purchase at the billing point, offering flexibility and control over the system operation.

4.4 – METHODOLOGY:

The methodology of this project focuses on developing a functional prototype of an Arduino-based Smart Shopping Trolley. The system is designed to automatically detect items placed in the trolley, compute the total bill in real-time, and notify both the shopper and the billing counter — thereby streamlining the shopping process and reducing checkout delays.

An RFID reader is used as the primary sensing element in this project. Each product is tagged with a unique RFID tag, and when the product is placed into or removed from the trolley, the RFID reader reads the tag and converts the tag's ID into an electrical signal. This tag-reading mechanism provides accurate and automated detection of items without manual scanning.

The Arduino UNO serves as the central control unit. It receives input data from the RFID reader, processes it using a program developed in the Arduino software environment, and executes the necessary logic to manage the shopping cart's contents. On detecting a valid RFID tag, the Arduino looks up the corresponding product details (e.g., name, price) from an internal database or stored lookup table. It then updates the current "cart list" and recalculates the total bill amount.

Once the bill is updated, the Arduino transmits the updated data to the output units. A display unit (such as an LCD or OLED) shows real-time billing information — including the product name, price, quantity (if multiple tags of the same item are supported), and cumulative total. This enables the shopper to continuously monitor their purchases and spending while shopping.

To ensure accuracy and prevent billing errors, the system continuously monitors for changes. If an item is removed from the trolley (e.g., the RFID tag is no longer detected in the read-zone), the Arduino updates the cart list, subtracts the corresponding item from the total, and refreshes the display. In case of any irregular detections (e.g., unregistered RFID tags), the system triggers an alert (e.g., via a buzzer or warning message on display) — thereby offering a safeguard against theft or misbilling.

By following this methodology, the Smart Shopping Trolley ensures accurate, automated item detection and real-time billing with minimal human intervention. The result is a more efficient shopping experience, significantly reduced queue times, fewer billing errors, and improved customer convenience..

CHAPTER - 5

5.1 HARDWARE ANALYSIS:

The hardware components form the physical structure of the Arduino-based Smart Shopping Trolley system. Each hardware element is selected to ensure accuracy, reliability and user-friendly operation.

RFID Reader (EM-18)
The RFID reader is the main sensing unit used to identify products. When a customer places an item in the trolley, the RFID reader scans the RFID tag attached to the product and sends the unique tag ID to the microcontroller. This enables quick and contactless product identification without manual entry.

RFID Tags
RFID tags are attached to each product in the store. Every tag contains a unique ID that represents the product's details such as name, price, and category. When scanned, the tag transmits this ID to the reader for billing purposes.

Arduino Uno / Arduino Nano
The Arduino board acts as the central processing unit. It receives data from the RFID reader, processes the product information, updates the total bill, and communicates with the display module. The microcontroller ensures smooth operation by managing inputs and outputs.

LCD Display / OLED Display
The display module shows the product name, product price, and the updated total bill when an item is added or removed. This gives customers a real-time view of their shopping details during trolley movement.

Buzzer
A buzzer is included to provide audio feedback for actions such as:
Successful product scan
Invalid/duplicate product scan
Item removed
This helps improve user interaction and system clarity.

Power Supply / Battery Pack
The system uses a rechargeable battery pack or power bank to run the Arduino board, LCD, RFID reader, and other components. Stable power ensures the trolley remains functional throughout the customer's shopping session.

5.2 COMPONENTS DESCRIPTION:

5.2.1 Arduino Nano

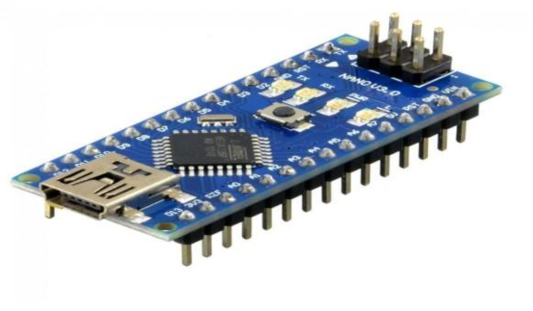


Figure 1.3 Arduino Nano

The Arduino Nano is a compact and powerful microcontroller development board widely used in embedded system projects that require small size and efficient performance. It is based on the ATmega328P microcontroller, which provides a perfect balance of simplicity, flexibility, and processing capability. Due to its small form factor and lightweight design, the Arduino Nano is highly suitable for portable and space-constrained applications such as the Smart Shopping Trolley.

The Nano board features 14 digital input/output pins, including 6 PWM outputs, 8 analog input pins, and operates at a clock speed of 16 MHz. These specifications make it capable of handling real-time tasks such as RFID tag detection, display updates, and buzzer control in the shopping trolley system.

The Arduino Nano includes a Mini-USB interface for programming and communication, allowing users to upload code easily using the Arduino IDE. It operates on a voltage range of 5V, and can be powered either through USB or via an external regulated power supply. The onboard voltage regulator, reset button, and ICSP header make it a versatile controller suitable for both basic and advanced embedded applications.

One of the major strengths of the Arduino Nano is its open-source hardware and software ecosystem, which supports a wide variety of libraries, sensors, and modules such as RFID readers, LCD displays, and buzzers. This makes the Nano an ideal choice for academic projects, automation systems, IoT prototypes, and hobby-level experiments. Its reliability, low power consumption, small size, and wide community support make the Arduino Nano a preferred platform for compact embedded designs such as the Smart Shopping Trolley system

5.2.2 RFID Reader (EM-18)

Figure 1.4: RFID Reader EM-18

The EM-18 RFID reader module is used to detect and read RFID tags attached to products. It operates at a frequency of 125 kHz and provides serial output to the microcontroller. When a product with an RFID tag is brought near the reader, the module reads the unique identification number (UID) and sends it to the Arduino.

The EM-18 features simple wiring, fast response, and reliable tag reading, making it ideal for automatic product identification. It requires a 5V power supply and communicates through UART, ensuring seamless data transfer to the microcontroller

5.2.3 RFID Tags

Figure 1.5: RFID Tags

RFID tags are attached to each product in the shopping area. Every tag contains a unique 12-digit code stored in its internal memory. When scanned, this code is sent to the Arduino, which matches it with the stored product database to identify the item.

These passive RFID tags do not require a power source and are low-cost, durable, and easy to attach, making them suitable for retail and billing automation applications.

5.2.4 LCD Display (16×2 LCD / OLED Display)

Figure 1.6: LCD Display

The LCD display is used to show real-time shopping information such as product name, price, quantity, and total amount. A 16×2 LCD is commonly used, which displays 32 characters, allowing clear visibility of billing details.

The LCD operates using the I2C module, reducing the number of pins required for communication with Arduino. It enhances user interaction by providing instant feedback for every scanned product.

5.2.5 Buzzer

Figure 1.7: Buzzer Module

A buzzer is included in the system to provide audio alerts during operations such as:

Successful product scan

Duplicate item scan

Product removal

Error notification

The buzzer improves the user experience by giving immediate sound feedback, ensuring smooth operation of the trolley system.

5.3 SOFTWARE ANALYSIS :

The software component of the Smart Shopping Trolley system is responsible for controlling system operations, item detection, billing calculation, and decision-making. It is developed using the Arduino Integrated Development Environment (IDE).

The Arduino program is written in Embedded C and includes logic to read digital data from the RFID reader. Each RFID tag value is processed and compared with the predefined product database stored in the Arduino memory. Whenever a tag is detected, the software checks whether the item is being added or removed and updates the cart contents accordingly.

Based on the tag identification, the software generates output signals to update the LCD display with the product name, price, quantity, and total bill amount. The program also ensures smooth item management by preventing duplicate entries and handling unexpected tag reads.

The software provides flexibility to modify parameters such as item database, price list, and buzzer alerts. This enables easy customization of the system to suit different shopping environments and requirements. Additional features such as invalid tag detection and bill reset functionality further enhance the system's usability.

Overall, the software plays a crucial role in ensuring reliable, automated, and efficient operation of the Smart Shopping Trolley system.

5.3.1 PROGRAM USED:

```
#include <LiquidCrystal_I2C.h> // Include the LCD library

#include <MFRC522.h> // Include the RFID library

#define SS_PIN 10

#define RST_PIN 9


LiquidCrystal_I2C lcd(0x27, 16, 2); // Change 0x27 if your LCD has a different I2C address

MFRC522 rfid(SS_PIN, RST_PIN); // Create MFRC522 instance


// Define product data (RFID UIDs and prices)

struct Product {

  String uid;

  String name;

  float price;

};

Product products[] = {

  {"2700875A344B", "Butter", 10.00},

  {"4000935ABAC5", "Milk", 20.00},

  {"0300234DA0F2", "Tea", 25.00}

};

float totalBill = 0.0;


void setup() {

  Serial.begin(9600); // Initialize serial communication

  SPI.begin(); // Initialize SPI bus

  rfid.PCD_Init(); // Initialize MFRC522
```

```
  lcd.init(); // Initialize the LCD
  lcd.backlight(); // Turn on LCD backlight
  lcd.setCursor(0, 0);
  lcd.print("Welcome to");
  lcd.setCursor(0, 1);
  lcd.print("Smart Trolley");
  delay(2000);
  lcd.clear();
}

void loop() {
  // Look for new cards
  if (rfid.PICC_IsNewCardPresent() && rfid.PICC_ReadCardSerial()) {
    String scannedUID = "";
    for (byte i = 0; i < rfid.uid.size; i++) {
      scannedUID += String(rfid.uid.uidByte[i], HEX);
    }
    scannedUID.toUpperCase(); // Convert to uppercase for comparison

    bool productFound = false;
    for (int i = 0; i < sizeof(products) / sizeof(products[0]); i++) {
      if (scannedUID == products[i].uid) {
        totalBill += products[i].price;
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print(products[i].name + " Added");
        lcd.setCursor(0, 1);
```

```
      lcd.print("Total: " + String(totalBill));
      productFound = true;
      break;
    }
  }

  if (!productFound) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Unknown Item");
    lcd.setCursor(0, 1);
    lcd.print("Scan Again");
  }

  rfid.PICC_HaltA(); // Halt PICC
  rfid.PCD_StopCrypto1(); // Stop encryption on PCD
  delay(2000); // Delay to prevent multiple scans
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Scan Next Item");
  lcd.setCursor(0, 1);
  lcd.print("Total: " + String(totalBill));
 }
}
```

CHAPTER – 6

RESULT :

The implemented smart shopping trolley system demonstrated effective performance in automating product identification, billing and price calculation inside a supermarket. During testing, the system successfully scanned products using RFID/barcode tags and automatically displayed the item name and price on the LCD. The Arduino based controller accurately processed the scanned product information and stored the purchased items into memory without human intervention.

A key outcome observed during testing was the significant reduction in manual billing time compared to the traditional cashier-based billing method. As every item was automatically detected inside the trolley itself, the need for long queues at the billing counter was considerably minimized. The application further prevented incorrect billing and manual entry errors by accurately reading product data through electronic sensing modules.

Another major observation was that the system worked reliably even when multiple items were scanned continuously. The Arduino Nano showed fast data processing and stable communication with RFID/barcode modules. Similarly, the buzzer indication and LCD output alert ensured proper confirmation of item addition or removal from the trolley.

The experimental operation highlighted easy usability where customers could directly see their running total amount, item list and final payable amount. The trolley also provided automatic calculation of total cost, which helped customers keep track of their budget while shopping.

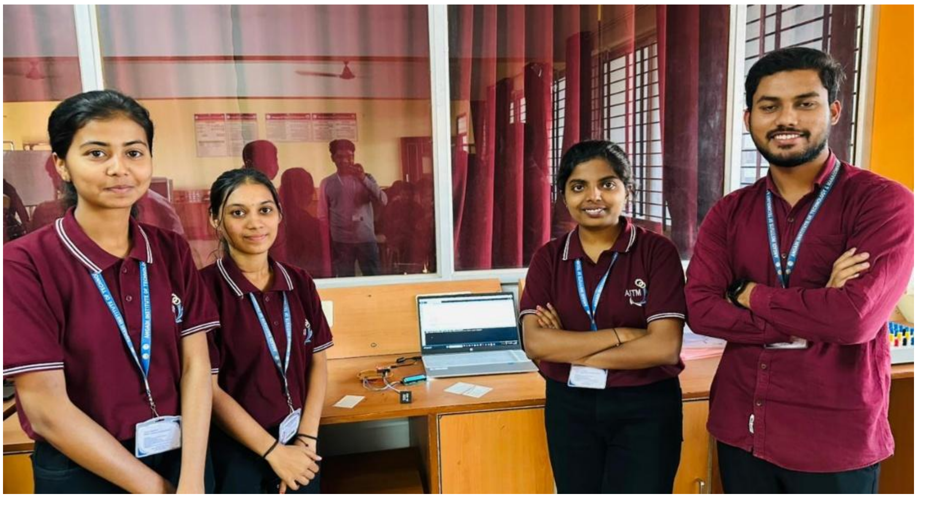


Figure 1.8: Demonstration of the Smart Shopping Trolley

The results also proved that the proposed trolley minimized crowding near counters and reduced customer waiting time. The system efficiently updated total price in real time while ensuring accurate billing even under continuous add–remove.

CHAPTER – 7

7.1 – APPLICATIONS:

The Arduino Nano–based Smart Shopping Trolley using RFID technology has a wide range of applications due to its ability to simplify the shopping process, reduce billing time, and provide a seamless automated experience for customers in retail environments.

1. The system is highly suitable for small and medium-scale supermarkets and retail stores where faster billing and customer convenience are essential. By using RFID-based product identification, the trolley automatically updates the cart total, reducing manual scanning and improving overall shopping efficiency.

2. The smart shopping trolley is extremely useful in crowded supermarkets and hypermarkets where long billing queues are a common issue. It provides customers with a dependable, self-assisted billing system that improves shopping speed and reduces waiting times.

3. The system can also be used in departmental stores and grocery outlets to improve inventory tracking and reduce human errors. Each product has an RFID tag, and as the customer adds or removes an item, the trolley updates the bill instantly, ensuring accuracy and preventing mismatches during checkout.

4. It is ideal for modern retail environments, shopping malls, and smart-store infrastructure in developing smart cities. The RFID-based automatic billing mechanism enhances customer experience and makes shopping more efficient, organized, and technologically advanced.

5. In large retail chains and supermarkets, the smart trolley system helps reduce manpower requirements during peak hours. It also increases the overall productivity of stores by minimizing manual scanning work and improving the effectiveness of product management.

6. The system is also applicable in wholesale markets and multi-product shopping centers for simplifying bulk item identification and billing. The enhanced accuracy of RFID technology helps store owners reduce billing mistakes, track product movement, and improve operational efficiency.

7.2 – ADVANTAGES:

The Arduino-based smart shopping trolley offers several advantages over conventional manual billing systems used in supermarkets. Its automated scanning capability improves shopping convenience and ensures accurate calculation of purchased items without human intervention.

1. One of the major advantages of this system is reduced billing time. By automatically detecting products through RFID/barcode scanning, the system avoids long queues and manual entry at counters, resulting in faster checkout compared to traditional billing methods.

2. The system operates automatically without the need for manual price entry. This reduces human effort and ensures consistent billing performance under varying product volumes and shopping conditions.

3. The use of an Arduino microcontroller makes the system cost-effective, compact and easy to modify. The design is flexible and can be expanded with additional modules such as wireless payment, mobile display, or central database connectivity according to future requirements.

4. The smart shopping trolley improves overall billing accuracy and minimizes pricing errors caused by human calculation. It helps in better utilization of automated retail systems and supports modern digital shopping environments.

5. Additionally, the system is reliable and suitable for operation in supermarkets, departmental stores, and shopping malls where manual billing takes significant time. Its simple design and low maintenance requirements make it ideal for practical use and educational retail automation projects.

7.3 – FUTURE ENHANCEMENT:

Future enhancements can be implemented in the Arduino-based Smart Shopping Trolley to improve its accuracy, efficiency, reliability, and user convenience. Some possible advanced improvements are listed below:

1. Integration of IoT (Internet of Things)

IoT connectivity can be added to send real-time billing details, product list, and trolley status to a cloud server or mobile app. This enables remote monitoring and helps store employees track trolley movement inside the mall.

2. RFID-Based Automatic Billing

High-accuracy RFID readers and long-range tags can be used to automatically detect products without manually scanning them. This enhancement reduces billing time and increases reliability.

3. Weight Sensor for Anti-Theft

A load cell/weight sensor can be installed to compare actual cart weight with RFID-detected items.

If a mismatch occurs, the system alerts the user and store staff, preventing shoplifting or billing errors.

4. Smart Navigation System

Ultrasonic sensors, IR sensors, or indoor GPS can be added to help the trolley avoid obstacles and navigate the

CHAPTER-8

CONCLUSION:

The Arduino-based Smart Shopping Trolley provides a practical and efficient solution for improving the shopping experience in supermarkets and retail stores. By automating product identification, billing, and cart monitoring using technologies such as RFID, load sensors, and Arduino microcontrollers, the system significantly reduces human effort and eliminates long billing queues. This enables customers to complete their shopping quickly and conveniently, demonstrating that even a simple embedded system can greatly enhance the efficiency of retail operations.

One of the key advantages of this project is its simplicity, reliability, and low cost. The use of basic components such as an Arduino board, RFID reader, LCD display, and buzzer makes the system easy to build, maintain, and customize for different store environments. Despite minimal hardware requirements, the trolley performs effectively by automatically detecting items, calculating the total bill, and notifying the customer about added or removed products. This shows that affordable embedded solutions can still deliver smooth and accurate performance in real-world applications.

The system's potential applications extend to supermarkets, retail shops, departmental stores, hypermarkets, and even libraries where automated item identification is needed. Its smart design helps reduce manpower, prevents billing errors, and enhances customer satisfaction. The system becomes especially useful in crowded stores, where traditional checkout counters lead to long waiting times and customer inconvenience.

Looking ahead, the smart trolley system offers several opportunities for improvement. IoT-based connectivity can be introduced to enable cloud billing, mobile-app integration, and real-time trolley tracking. Additional enhancements may include the use of weight sensors for anti-theft protection, advanced RFID modules for long-range detection, voice-based assistance for easy user interaction, and machine-learning algorithms capable of recommending products based on shopping history. These advancements can make the trolley even more intelligent, secure, and user-friendly.

In summary, the Arduino-based Smart Shopping Trolley successfully demonstrates that automation can significantly improve the retail shopping experience. Its efficiency, affordability, and potential for future development make it a promising solution for modernizing billing systems and enhancing customer convenience in supermarkets and retail environments.

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